Project Overview: Credit Card Weekly Dashboard

1. Approach

This project was developed to analyze customer profiles and credit card transaction behavior using Power BI. Two interactive dashboards were created:

- **Customer Report Dashboard**: Focused on customer demographics, segmentation, and satisfaction metrics.
- **Transaction Report Dashboard**: Focused on revenue, transaction patterns, and week-over-week performance.

Data from both customer and transaction datasets were cleaned, transformed, and modeled using relationships based on Client_Num.

DAX calculations were implemented to derive:

- Age Groups and Income Groups (Low, Medium, High)
- Total Revenue and Transaction Metrics
- Current Week Revenue and Previous Week Revenue
- Week-over-Week (WoW) change analysis

Interactive slicers were added for Gender, Age Group, Income Group, Card Category and Week Number to enable dynamic filtering across visuals.

2. Key Insights

- Customer distribution varies significantly across income and age groups, indicating different financial segments.
- Spending behavior is concentrated on bills, fuel and entertainment, highlighting key revenue drivers.
- Week-over-week analysis shows fluctuations in revenue, influenced by missing weeks in the dataset.
- Customers with higher income levels tend to have higher credit limits.
- Satisfaction scores provide insight into customer experience across segments.

3. Challenges & Solutions

One key challenge encountered was implementing the **drill-through functionality**.

- Initially, drill-through was not working because the required field (Client_Num) was not properly integrated into the main dashboard visuals.

- This was resolved by ensuring that visuals used for drill-through included Client_Num, enabling proper context transfer between pages.
- Additionally, understanding how Power BI handles filter context was essential in making the drill-through functional.

Another challenge was handling **missing week data** in the transaction dataset:

- Some weeks were not continuous, which affected week-over-week calculations.
- This was resolved by modifying DAX logic to dynamically identify the previously available week instead of assuming sequential week numbers.

4. Interactivity & Design

The dashboards were designed to be:

- Clean and visually intuitive
- Interactive using slicers and filters
- Structured to provide both high-level insights and detailed drill-down capabilities

Users can:

- Filter by Gender, Age Group, and Income Group
- Analyze trends across different time periods
- Drill through to view detailed customer-level information